- Faithful Error Propagation (Y_{prop}): 197 / 200 (98.50%) adopted the corrupted value E and correctly executed the downstream multiplication $E \times 4$.
- Attention-Slip Divergence (Y_{div}): Exactly 3 / 200 (1.50%) confused the unit price with the hourly rate.

E. Finite-Sample Exact Confidence Bounds

Because asymptotic inference collapses at zero events ($k = 0$), we compute exact Clopper-Pearson binomial confidence bounds [12]:

$$p_U = 1 - (\alpha/2)^{1/n}, \quad p_{U,\text{one-sided}} = 1 - \alpha^{1/n}$$

As reported in Table III, assuming 200 independent trials yields a two-sided 95% interval of $[0.000\%, 1.828\%]$ (one-sided upper bound: 1.487%). However, because rollouts within problem clusters share prefixes, treating the 20 problem clusters as the sampling unit ($C = 20$) provides a conservative cluster-level sensitivity bound of $\leq 13.911\%$ (two-sided interval: $[0.000\%, 16.843\%]$).

TABLE III
Clopper-Pearson Exact Confidence Bounds for Zero Observed Recovery ($k = 0$)

Aggregation Tier	Effective N	One-Sided 95% Upper	Two-Sided 95% CI
Rollout Level (Total)	200	1.49%	[0.00%, 1.83%]
Model Level (Per Checkpoint)	100	2.95%	[0.00%, 3.62%]
Cluster Level (Sensitivity)	20	13.91%	[**0.00%**, **16.84%**]

VI. Discussion: Mechanistic Nature of Continuation Inertia

Why do small language models exhibit near-total continuation inertia? Autoregressive language models are trained via next-token prediction to model the joint distribution $P(x_t \mid x_{<t})$. When presented with a prompt prefix containing an arithmetic error, the attention mechanism treats all prefix tokens as grounded context. In standard pretraining and instruction tuning, the model is penalized for contradicting prompt premises. Consequently, the model's most probable trajectory is to execute subsequent derivations that are logically consistent with the preceding tokens, even when those tokens contain an objective arithmetic flaw.

This demonstrates that without explicit verification prompting or external feedback tools (such as CRITIC or Reflexion), autoregressive models function as faithful continuation engines rather than autonomous error detectors.

VII. Limitations

- Model Scale: Evaluated on two 1.5B checkpoints; larger models (7B–72B) or reasoning models with internal test-time search (e.g., DeepSeek-R1) may exhibit different behaviors.
- Task Domain: Restricted to 20 elementary arithmetic rate-multiplication problems; does not evaluate open-domain theorem proving or multi-branch logic.
- Decoding Scaffolding: Evaluated under unprompted continuation without reflective scaffolding (*"check your work"*) or tool interaction.
- Observational Models: Checkpoint comparisons remain observational due to chat template differences.
- Statistical Power: With $C = 20$ problem clusters, the study has low power to detect subtle recovery rates ($|D| < 0.10$).
- Human Validation: Internal blinded human audit of 40 rollouts showed 100% concordance with structural parsing, but multi-rater crowdsourced adjudication remains pending future work.

VIII. Reproducibility & Open Source

All experimental materials, code, and neural rollout logs are permanently open-sourced at https://github.com/shamddd/recovery_eval. To reproduce all statistical tables in under 1 second:

```
git clone https://github.com/shamddd/recovery_eval.git
cd recovery_eval
python3 reproduce_results.py
python3 -m pytest tests/ -v
```

IX. Conclusion

We introduced RecoveryEval, an open-source evaluation framework for language model reasoning trajectories. By eliminating control answer leakage, 64-token truncation censoring, and parser artifacts, we demonstrated that the historical negative interaction was an evaluation artifact. Small language models exhibit near-total continuation inertia (98.5%), faithfully propagating injected errors rather than self-correcting. Our findings highlight the necessity of state-matched, provenance-aware infrastructure for reasoning diagnostics.

Declarations & AI Assistance Disclosure

In the preparation of this manuscript, the author utilized automated AI research tools (including Google Antigravity and Large Language Models) to assist with automated test suite refactoring, simulated adversarial peer review critiques, and LaTeX formatting. All primary neural rollouts were generated on independent GPU hardware (RunPod RTX 4090) using official checkpoints, and all statistical counts, exact bounds, and interpretations were verified against raw logs. The human author takes full scientific responsibility for the work. In compliance with publishing standards, no AI system is listed as an author or co-author.

References

[1] K. Cobbe, V. Kosaraju, M. Bavarian, M. Chen, H. Jun, L. Kaiser, M. Plappert, J. Tworek, J. Hilton, R. Nakano, C. Hesse, and J. Schulman, "Training verifiers to solve math word problems," in arXiv preprint arXiv:2110.14168, 2021.

[2] A. Yang, B. Zhang, B. Zheng, D. Liu, J. Zhou et al., "Qwen2.5-Math technical report: Toward open math large language models with mathematical reasoning capabilities," arXiv preprint arXiv:2409.12122, 2024.